

Team 13: Cost-Aware Adaptive Retrieval for Enterprise Support

1. Project Idea and Objectives

This **application-focused project** will build a retrieval assistant for enterprise support that searches a company's internal documents and adjusts its effort based on the information needed to answer a question. Simple questions may be answered using previously verified responses, while complex questions (novel or multi-part) may require searching and combining information from several documents. If the evidence is weak, another round of agentic requerying will happen. If no query finds sufficient evidence, the system refuses to answer rather than guess. We are planning on testing it using both simple and difficult support questions to measure the accuracy and reliability of its answers. Our main goal is to reduce average response time and operating cost without reducing the answer quality.

2. Related Work

Enterprise support tools today generally use Retrieval Augmented Generation (RAG) with a single fixed retrieval strategy applied uniformly to all the queries regardless of their level of difficulty. This wastes computation on easy and repeated questions. Adaptive RAG (Jeong et al., 2024) selects different search strategies according to question. Self RAG (Asai et al., 2024) teaches a model to decide when to search and assess its responses. Neither study did not fully cover its latency or cost inside an end to end pipeline, and touched the heterogeneous, noisy, jargon heavy documents found inside a company. We instead evaluate a realistic benchmark of documents across several source types such as, Slack, and enterprise wikis, intended for industrial implementation.

3. Novelty and Impact

Our contribution is to apply and evaluate adaptive retrieval routing, informed by adaptive RAG in an enterprise support setting, combining reuse of naive approaches, and quantifying a second agentic search when needed, and refusal to answer without sufficient evidence. We also plan to optimize each stage of the pipeline by teaching the search model which documents to prioritize and testing methods that quantify which approaches improve search speed and accuracy. We expect that avoiding repeated work on previously answered questions could save time and money, while additional searching could improve responses to difficult questions. This could help support teams answer routine questions more efficiently. This could be implemented in enterprise support to manage both internal and customer relations.

4. Method and Evaluation

We will use a manageable portion of ONYX EnterpriseRAG-Bench, a benchmark for answering questions using company documents. Our solution retrieves candidate passages/chunks using naive keyword search, reranks them for relevance, and routes each question through one of three paths: cache lookup for repeat questions, single-pass retrieval and generation for novel but answerable questions, or a bounded second retrieval round for low-confidence cases, with explicit refusal when evidence remains insufficient. We aim to prepare approximately 300 questions or more, covering straightforward lookups, questions requiring information from multiple documents.

We will measure answer correctness using precision@k, faithfulness using retrieval of relevant evidence, mean response time, and mean cost per correctly answered questions for each case. Initial dataset partition will be 70% development set, 15% validation set and 15% held-out test set. Our six-person team will divide data preparation, document search, decision logic, evaluation, integration, and analysis of metrics over ten weeks.

5. Resources

We plan to use local computers, and if required for fine tuning a re-ranker or a model on specific domains, we will use the help of CARC.

6. Risks

The two main risks here are the cache serving wrong context answers to similar but different questions and the system treating weak retrieved evidence as sufficient. If adaptive search does not improve performance, we will retain the strongest fixed approach and report our findings.

7. Milestones

Milestone 1, by the end of **Week 4**: Prepare and load the documents in the respective Vector DB, NoSql DB, retrieval and reranker implement basic search methods, and complete an initial evaluation.

Milestone 2, by the end of **Week 10**: Complete the adaptive agent, compare approaches, review responses, and deliver an understanding of how well and efficient agents can retrieve the documents compared to our naive approaches. with a final report on when adaptive routing wins and when it does not.